

Dhruv Rajput

dhruvrajput@email.com | github.com/DHRUVRAJPUTTT | linkedin.com/in/dhruv-rajput2007 | Patiala, Punjab

EDUCATION

Thapar Institute of Engineering and Technology

2024 – 2028

Bachelor of Technology (B.Tech), Computer Science

Patiala, Punjab

- **Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Computer Networks, Operating Systems

PROJECTS

Text-to-SQL Query Engine

Python, FastAPI, React, OpenAI API, ChromaDB, LangChain

- Built an end-to-end RAG pipeline embedding database schemas into a ChromaDB vector store and retrieving only the **top-3 relevant tables** per query, eliminating context-overflow issues common in naive text-to-SQL systems
- Engineered a semantic layer annotating every column with business-context descriptions to guide the LLM toward accurate joins, aggregations, and filters
- Implemented a **HITL safety guard** blocking non-SELECT SQL (INSERT, UPDATE, DELETE, DROP) and requiring explicit user confirmation before execution
- Built a React + TypeScript frontend with syntax-highlighted SQL display, a pageable results table, and a schema explorer, connected to a FastAPI backend with observability logging

Enterprise Agentic RAG Pipeline

Python, LangGraph, FastAPI, Qdrant, Portkey

- Built a **production-grade agentic RAG system** with LangGraph for cyclic reasoning, multi-step planning, and conversation memory, distinguishing true technical data from noisy inputs via semantic re-ranking
- Implemented **NeMo Guardrails** to block off-topic, jailbreak, and injection inputs before retrieval, and routed all LLM calls through a **Portkey gateway** with automatic fallback between primary and backup Groq keys
- Built an enterprise search layer using **Qdrant Cloud** vector search, 3072-dim Gemini embeddings, and FlashRank local reranking for high-performance retrieval
- Developed a **RAGAS-powered evaluation suite** (6 metrics) with full observability via Pydantic Logfire and LangSmith tracing across every agent node

Electricity Prediction AI

Python, Machine Learning

- Built a machine learning model to predict electricity consumption patterns using historical time-series data with regression and time-series analysis techniques
- Implemented data preprocessing pipelines, feature engineering, and model evaluation metrics to improve forecasting accuracy
- Visualized prediction results and deployed the model for practical energy demand forecasting use cases

Network Traffic Classification

Python, Cybersecurity, ML

- Designed and trained a machine learning model to classify network traffic for cybersecurity threat detection
- Analyzed packet-level and flow-level features to distinguish between normal and malicious traffic patterns
- Applied supervised learning algorithms and feature engineering, bridging Computer Networks and ML concepts to address real-world security challenges

SKILLS

Languages : Java, C++, C, Python, Go (Golang)

AI / ML : LangChain, LangGraph, RAG, OpenAI API, ChromaDB, Qdrant, Prompt Engineering

Web & APIs : FastAPI, React, TypeScript, REST APIs

DevOps & Tools : Docker, CI/CD Pipelines, Git, GitHub

Core CS : Data Structures & Algorithms, OOP, Computer Networks, Operating Systems